

Natural properties

Advanced Introduction to Metaphysics
7 December 2020

1. Intuitive examples

2. Perfect naturalness and degrees of naturalness

- Perfect naturalness as being maximally natural
- The length-of-definition criterion

3. Connections between naturalness and other (easier) notions

(i) Similarity

a) General similarity

- 'Sharing of natural properties makes for similarity; division by natural properties making for dissimilarity'
- Defining naturalness in terms of degrees of similarity
- Defining similarity in terms of naturalness: a suggestion

Division: the degree of dissimilarity between x and y is the sum of the degrees of naturalness of all the properties that divide x and y .

b) Duplication and intrinsicness

Duplication: x is a duplicate of y if and only if there is a one-to-one, onto function f from the parts of x to the parts of y , such that for any perfectly natural n -ary property R and any finite sequence $\langle x_1, \dots, x_n \rangle$ of parts of x , $Rx_1 \dots x_n$ just in case $Rf(x_1) \dots f(x_n)$.

- Lewis: a property is *intrinsic* iff it never divides duplicates.

c) Qualitative indiscernibility and qualitateness

Indiscernibility: x is qualitatively indiscernible from y if and only if there is a one-to-one, onto function f from all objects x to all objects y , such that $f(x)=y$ and for any perfectly natural n -ary property R and any finite sequence $\langle x_1, \dots, x_n \rangle$ of parts of x , $Rx_1 \dots x_n$ just in case $Rf(x_1) \dots f(x_n)$.

Qualitative Supervenience: If f is a universally defined bijection such that for every perfectly natural n -ary property R and objects $x_1, \dots, x_n$, $Rx_1 \dots x_n$ at w iff $Rf(x_1) \dots f(x_n)$ at w' , then for every qualitative m -ary property S and objects $x_1, \dots, x_m$, $Sx_1 \dots x_m$ at w iff $Sf(x_1) \dots f(x_m)$ at w' .

(ii) Modality

a) Completeness

TSB: Necessarily, every true proposition is metaphysically necessitated by the conjunction of (a) all true propositions $Rx_1...x_n$ or $\neg Rx_1...x_n$, where R is a perfectly natural n -ary property, and (b) the proposition that every object is D , where D is the disjunction of the haecceities of all objects.

b) Independence

- Minimality: Qualitative Supervenience or TSB becomes false if we replace ‘perfectly natural property’ with ‘perfectly natural property distinct from R ’ for any given perfectly natural R .
 - Worries: converse relations;
- Lewis’s principle of recombination
- More ambitious independence claims

Combinatorialism: If $F_1 \dots F_n$ are perfectly natural and R is a possibly instantiated *pattern-specifying* higher order relation of the appropriate type, then possibly $RF_1...F_n$.

(iii) Laws of nature

BSA: p is law of nature iff [p is a generalization and] p belongs to the set of propositions closed under metaphysical necessitation that has the highest combined score for *simplicity* [$\approx$ naturalness] and *strength*.

(iv) Mental and linguistic content (magnetism)

- Lewis’s toy “use-plus-eligibility” theory.
- Lewis’s theory

(Dorr and Hawthorne’s reconstruction:) Step one: analyse ‘ a has credence function C and value function C' ’ as ‘the three-place relation that plays the credence-value role for a ’s species holds between a , C and V ’. Step two: analyse ‘ R plays the credence-value role for species s ’ as a matter of R ’s achieving an optimal balance of several desiderata, one of which is that it should not be too common for members of s to perform actions that are not optimal according to the C and V to which they are mapped by R , and another of which is that R itself should not be too unnatural.

- ‘The more natural a property is, the easier it is to refer to’
 - Maybe that should be ‘refer determinately’ or ‘refer uniquely’
 - Spelling out in terms of physics-driven measures

(v) Epistemology

a) Ampliative inference

Bayesian implementation: A probability function is a prior credence function iff it is a weighted sum of all sufficiently natural probability functions in which the weight of each function is (roughly) proportional to its naturalness.

b) Epistemic value

Multiplicative Value Naturalness: When a proposition is true, believing (knowing) it is good in proportion to its naturalness; when it is false, believing it is bad in proportion to its naturalness.

4. Bad debates about naturalness

5. Debate: how much of the theoretical role is satisfied?

Example: what should we say about *determinate mass properties* like *having a mass of 4.183748 grams*? Different roles arguably pull in different directions.

- Saying they are perfectly natural makes trouble for Independence; some versions of the connection to similarity such as *Division*; *Laws*; and *Magnetism*. Similar remarks apply to mass-interval properties.
- Saying that they are *not* perfectly natural makes for trouble if we hold that *objects with different masses cannot be duplicates*. (But we might contemplate giving that up!)

Example: what should we say about moral properties like *being morally permissible*?

- Saying that they are highly unnatural makes trouble for Magnetism given the popular view that they are quite easy to refer to.
- Saying that they are perfectly natural makes potential trouble for Independence.
- Saying that they are pretty natural but not perfectly natural makes potential trouble for the length-of-definition criterion

6. Debate: how vague is 'natural'?